6/5/2025(M)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: Jan 2025– May 2025		
Maximum Marks: 50 Examination: End-Semester Examination		Duration: 2 Hrs. +40min
Programme code: 01 Programme: B. Tech. (Computer Engineering)	Class: LY	Semester: VIII
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP/~~ETRX/EXTC/IT/MECH/RAI/AIDS/CCE/EXCP~~	
Course Code: 116U01E853	Name of the Course: Internet of Everything PWD	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **Ten out of Fifteen**	10
i.	**What does the logical design of IoT primarily define?** A. The shape and size of IoT devices B. Hardware layout of IoT systems C. The functional blocks and communication models D. Power requirements of sensors	01
ii.	**Which of the following is not an IoT enabling technology?** A. Wireless Sensor Networks B. Blockchain C. Cloud Computing D. 3D Printing	01
iii.	**Which communication model in logical design supports asynchronous communication between devices?** A. Request-Response B. Publish-Subscribe C. Client-Server D. Peer-to-Peer	01
iv.	**Which of the following is not a component of IoT logical design?** A. Functional blocks B. Communication models C. Deployment locations D. Communication APIs	01
v.	**Which technology is most commonly used for communication in home automation?** A. GSM B. Zigbee C. Satellite D. Li-Fi	01

vi.	**Smart grids in energy management help in:** A. Increasing power consumption B. Real-time monitoring and demand response C. Playing music in power stations D. Reducing communication speed	01
vii.	**A key difference between IoT and M2M is that IoT:** A. Does not use the internet B. Only supports wired communication C. Integrates cloud computing and big data analytics D. Works only with home devices	01
viii.	**What is the primary purpose of Software Defined Networking (SDN) in IoT?** A. Controlling physical sensors B. Separating control plane from data plane to allow centralized network management C. Enhancing sensor battery life D. Encrypting sensor data	01
ix.	**Which of the following is the main microcontroller used in Arduino Uno?** A. ATmega328P B. PIC16F877A C. ESP8266 D. ARM Cortex-M3	01
x.	**Which module is commonly used to connect an Arduino to the Internet via Wi-Fi?** A. Bluetooth HC-05 B. ESP8266 C. GSM SIM800L D. Zigbee XBee	01
xi.	**In an IoT-enabled supply chain system, what is typically tracked using sensors?** A. Employee salaries B. Product location and condition C. Marketing trends D. Cloud costs	01
xii.	**In IoT-cloud architecture, data is typically sent to the cloud for:** A. Display only B. Archival only C. Processing, storage, and analytics D. Reducing network size	01

xiii.	**What is the first step in a generic IoT design methodology?** A. Deploying devices B. Defining use-case and requirements C. Selecting cloud storage D. Coding the app	01
xiv.	**What is the main goal of the system integration phase in IoT design methodology?** A. Run simulations B. Purchase devices C. Combine hardware and software components to work together D. Prepare marketing campaigns	01
xv.	**In Industrial IoT, digital twins are used for:** A. Data duplication B. Real-time simulation and monitoring of physical systems C. User authentication D. Reducing sensor count	01
Q.2	Attempt any **one**	
i)	a) What do you understand by Network Function Virtualisation? b) With a diagram, explain the architecture of the same. Also mention the key elements of the same.	10 05 05
ii)	What are the special features of M2M with respect to IoT? Explain M2M architecture in detail.	05 05
Q.3	Attempt any **one**	
i)	Explain briefly following functions with reference to Arduino Programming giving example with appropriate syntax: • digitalWrite() • analogRead() • delay() • loop() • setup()	10 02 02 02 02 02
ii)	What is considered as physical model of IoE system? Explain in Detail (10)	
Q.4	Attempt the following	
	a) With example explain IoT level 2 implementation b) In which type of application IoT level 5 is to be implemented, explain?	10 05 05
Q.5	Attempt the following	
i.	Explain any 5 main challenges faced while using IoT in business process.	10
ii.	Explain merits of Edge Analytics in IoT.	05
iii.	What are the components of IoT Data Analytics?	05